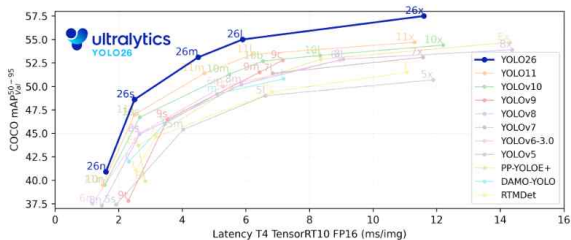


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Ultralytics YOLO26

Overview

[Ultralytics](#) YOLO26 is the latest evolution in the YOLO series of real-time object detectors, engineered from the ground up for **edge and low-power devices**. It introduces a streamlined design that removes unnecessary complexity while integrating targeted innovations to deliver faster, lighter, and more accessible deployment.



On this page

[Overview](#)[Key Features](#)[Supported Tasks and Modes](#)[Performance Metrics](#)[Usage Examples](#)[YOLOE-26: Open-Vocabulary Instance Segmentation](#)[Usage Example](#)[Citations and Acknowledgments](#)

FAQ

What are the key improvements in YOLO26 compared to YOLO11?

What tasks does YOLO26 support?

Why is YOLO26 optimized for edge deployment?

How do I get started with

Detect

Segment

Classify

Pose

OBB

Object Detection



Object detection is a task that involves identifying the location and class of objects in an image or video stream.

The output of an object detector is a set of bounding boxes that enclose the objects in the image, along with class labels and confidence scores for each box. Object detection is a good choice when you need to identify objects of interest in a scene, but don't need to know exactly where the object is or its exact shape.

On this page

Models

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Dataset format

Val

Predict

Export

FAQ

Can I train and deploy detection models without coding?

How do I train a YOLO26 model on my custom dataset?

What pretrained models are available in YOLO26?

How can I validate the accuracy of my trained YOLO model?

What formats can I export a YOLO26 model to?



main

344 Branches 659 Tags

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Add file

<> Code



lakshanthad and Laughing-q

Migrate benchmarks CI from ubuntu-latest to cpu-...



235ebb3 · 1 hour ago

3,879 Commits

.github	Migrate benchmarks CI from ubuntu-latest to cpu-latest (#2...	1 hour ago
docker	Fix torch and torchvision versions on JetPack 5 Docker (#241...	last week
docs	Add NVIDIA DALI GPU preprocessing guide (#24102)	12 hours ago
examples	Fix redirected URLs (#24255)	5 days ago
tests	ultralytics 8.4.39 Use dashed incremented run paths (#241...	2 days ago
ultralytics	ultralytics 8.4.40 Per-image Precision and Recall (#24089)	18 hours ago
.dockerignore	Create .dockerignore (#18534)	2 years ago
.gitignore	ultralytics 8.3.243 Deduplicate ConsoleLogger progress ba...	4 months ago
CITATION.cff	Update extrajs with 0.3.20/dist/embed.min.js (#20230)	last year
CONTRIBUTING.md	Fix redirected URLs (#24255)	5 days ago
LICENSE	Update LICENSE to AGPL-3.0 (#2031)	3 years ago
README.md	Fix redirected URLs (#24255)	5 days ago
README.zh-CN.md	Fix redirected URLs (#24255)	5 days ago
mkdocs.yml	Add NVIDIA DALI GPU preprocessing guide (#24102)	12 hours ago
pyproject.toml	ultralytics 8.4.25 TensorFlow.js fix pin yaf<0.13.0 (#23985)	last month

About

Ultralytics YOLO 🚀

docs.ultralytics.com

python cli tracking
machine-learning computer-vision
deep-learning hub pytorch yolo
image-classification object-detection
pose-estimation instance-segmentation
ultralytics rotated-object-detection
yolov8 segment-anything
yolo-world yolo11 yolo26

Readme

AGPL-3.0 license

Code of conduct

Contributing

Security policy

Cite this repository

Activity

Custom properties

56.2k stars

255 watching

10.8k forks

Report repository

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Python

```
from ultralytics import YOLO

# Create a new YOLO model from scratch
model = YOLO("yolo26n.yaml")

# Load a pretrained YOLO model (recommended for training)
model = YOLO("yolo26n.pt")

# Train the model using the 'coco8.yaml' dataset for 3 epochs
results = model.train(data="coco8.yaml", epochs=3)

# Evaluate the model's performance on the validation set
results = model.val()

# Perform object detection on an image using the model
results = model("https://ultralalytics.com/images/bus.jpg")

# Export the model to ONNX format
success = model.export(format="onnx")
```

Train

Train mode is used for training a YOLO model on a custom dataset. In this mode, the model is trained using the specified dataset and hyperparameters. The training process involves optimizing the model's parameters so that it can accurately predict the classes and locations of objects in an image.

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Train

On this page

[Train](#)[Val](#)[Predict](#)[Export](#)[Track](#)[Benchmark](#)[Using Trainers](#)[FAQ](#)

How can I integrate YOLO into my Python project for object detection?

What are the different modes available in YOLO?

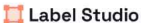
How do I train a custom YOLO model using my dataset?

How do I export YOLO models for deployment?

Can I validate my YOLO model on different datasets?

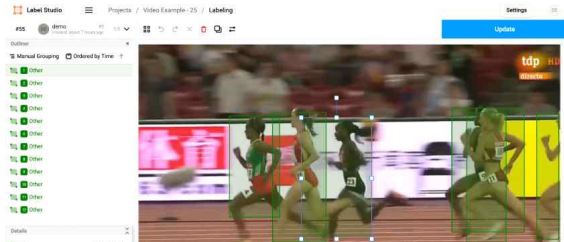


Labeling is now part of the Label Studio community!



Label Studio is a modern, multi-modal data annotation tool

Labeling, the popular image annotation tool created by Tzatalin with the help of dozens contributors, is no longer actively being developed and has become part of the Label Studio community. Check out [Label Studio](#), the most flexible open source data labeling tool for images, text, hypertext, audio, video and time-series data. [Install](#) Label Studio and join the [slack community](#) to get started.



Used by 2k



Contributors 97

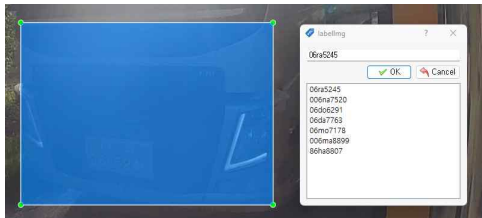
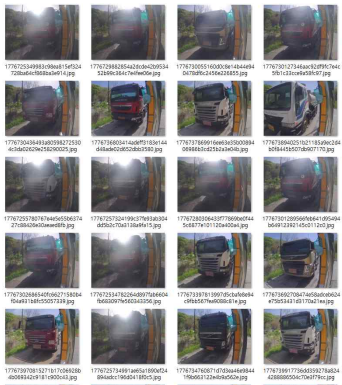


+ 83 contributors

Languages



객체 라벨링



대서산업 트럭 인식을 위해 해당 이미지들과 같이 정면으로 찍힌 사진들을 대상으로 라벨링 실시

라벨링을 하는 툴을 사용하여
드래그된 부분을 미리 정해둔 라벨 목록으로 지정
(한글 라벨링이 되지 않아 영어로 변경)

학습 코드

The image shows a code editor interface with a sidebar on the left displaying a file tree and a main editor area on the right showing a Python script. The sidebar lists files including 'YOLO', 'coco8', 'runs', 'test_img', 'venv', and various image files. The main editor shows the 'test.py' file with the following code:

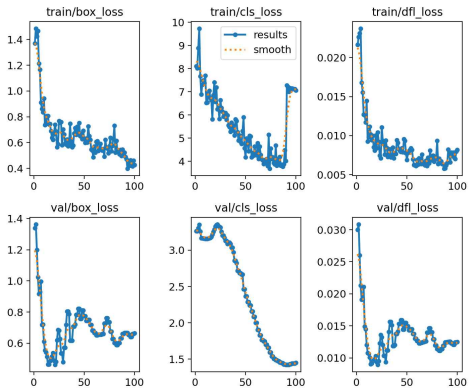
```
1 from ultralytics import YOLO
2
3 def main():
4     # Load a pretrained YOLO26n model
5     model = YOLO("yolo26n.pt")
6
7     # Train the model
8     # Windows에서는 workers=0으로 설정하면 더 안정적일 수 있으나,
9     # 아래 main 블록만 잘 감싸줘도 대부분 해결됩니다.
10    train_results = model.train(
11        data="test.yaml",
12        epochs=100,
13        imgsz=640,
14        device="cuda", # 이제 GPU(CUDA)를 사용합니다!
15    )
16
17    # Evaluate performance
18    metrics = model.val()
19
20    # Perform detection
21    results = model[Any]("17767437240729769394819a04d1dac7728b4eb3478f0.jpg")
22    results[0].show()
23
24    # Export to ONNX
25    path = model.export(format="onnx")
26
27 if __name__ == '__main__':
28     main()
```

학습 과정

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
94/100	2.5G	0.4427	7.164	0.00785	14	640: 100%	2/2 5.4it/s 0.4s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	15	15	0.844	0.769	0.879	0.764 1/1 8.8it/s 0.1s
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
95/100	2.5G	0.4217	7.09	0.007469	14	640: 100%	2/2 6.5it/s 0.3s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	15	15	0.844	0.769	0.879	0.764 1/1 8.1it/s 0.1s
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
96/100	2.5G	0.4621	7.149	0.007848	14	640: 100%	2/2 5.4it/s 0.4s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	15	15	0.846	0.765	0.879	0.764 1/1 7.4it/s 0.1s
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
97/100	2.5G	0.4421	7.13	0.007204	14	640: 100%	2/2 5.6it/s 0.4s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	15	15	0.846	0.765	0.879	0.764 1/1 11.0it/s 0.1s
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
98/100	2.5G	0.4155	7.119	0.007038	14	640: 100%	2/2 5.9it/s 0.3s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	15	15	0.831	0.77	0.885	0.766 1/1 9.2it/s 0.1s
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
99/100	2.5G	0.4592	7.12	0.007977	14	640: 100%	2/2 6.8it/s 0.3s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	15	15	0.831	0.77	0.885	0.766 1/1 10.9it/s 0.1s
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
100/100	2.5G	0.4262	7.066	0.008202	14	640: 100%	2/2 5.9it/s 0.3s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	15	15	0.82	0.779	0.885	0.766 1/1 11.4it/s 0.1s

객체 탐지모델 학습

Yolo26 모델을 사용하여 라벨링된 데이터 30개를 학습시킨 후 인식 테스트



현재 수집된 데이터 : 2023.11.16 ~ 현재까지 (122,450개 파일)

